

REVIEW 1

Data Understanding & Analysis

B.Tech First Year | MDSA | Python

1. Title

Exploratory Data Analysis on a Dataset Using Python

2. Problem Statement

Raw datasets are messy. They have missing values, duplicate rows, and wrong data types. Without cleaning the data first, our analysis gives wrong results. This project loads, cleans, and explores a dataset using Python to find useful information from it.

3. Objectives

- Load a dataset from a CSV file and view its basic details.
- Clean the data by fixing missing values and removing duplicate rows.
- Fix wrong data types so the data is ready for analysis.
- Find basic statistics like mean, median, and standard deviation.
- Understand the data and write simple conclusions from it.

4. Dataset Description

Source

Download any dataset from Kaggle (www.kaggle.com) or UCI ML Repository and save as a CSV file.

Features (Columns)

Column Type	Examples	Meaning
Numerical	age, marks, salary	Numbers like 25, 85.5
Categorical	gender, city, dept	Labels like Male, CSE
Date / Time	date, timestamp	Dates like 2024-01-15
Target	result, grade, price	Main column we study

Size

Can be 100 rows (small) to millions of rows (large). The code below works for any size.

```
import pandas as pd

import numpy as np
```

```
# Step 1: Read the data

df = pd.read_csv('dataset.csv')

# Step 2: Check shape (rows, columns)

print(df.shape)

# Step 3: See first 5 rows

print(df.head())
```

5. Data Cleaning

Missing Values

Missing values show as NaN. Fill number columns with median, text columns with mode.

```
# Check missing values

print(df.isnull().sum())

# Fill number columns with median

num_cols = df.select_dtypes(include='number').columns

df[num_cols] = df[num_cols].fillna(df[num_cols].median())

# Fill text columns with most common value

cat_cols = df.select_dtypes(include='object').columns

for col in cat_cols:

    df[col].fillna(df[col].mode()[0], inplace=True)
```

Duplicates

Duplicate rows are repeated entries. Remove them to keep the data clean.

```
# Check duplicates

print(df.duplicated().sum())
```

```
# Remove duplicates

df.drop_duplicates(inplace=True)

df.reset_index(drop=True, inplace=True)
```

Data Correction

Fix wrong data types so numbers and dates are stored correctly.

```
# Fix number column

df['price'] = pd.to_numeric(df['price'], errors='coerce')

# Fix date column

df['date'] = pd.to_datetime(df['date'], errors='coerce')

# Clean text column

df['name'] = df['name'].str.strip().str.title()

print(df.dtypes)
```

6. Exploratory Data Analysis (EDA) — Basic Stats

EDA helps us understand the data — its averages, spread, and unique values.

```
# Full summary (mean, min, max, std ...)

print(df.describe())

# Dataset info (column types, null counts)

print(df.info())

# Mean, Median, Std of number columns

print(df[num_cols].mean())

print(df[num_cols].median())
```

```
print(df[num_cols].std())

# Count of each value in text columns

for col in cat_cols:
    print(df[col].value_counts())
```